

ESP32 Lightweight Multimodal NIDS

2026.08.06

Bottom line

1. **Pipeline works** — labeled 100 ms windows → train → on-device tree export.
2. **Multimodal helps** — full vs NIDS-only: $\Delta F1 \approx +0.27$.
3. **But almost all of that gain is free-heap** (importance ≈ 0.88 ; drop heap → F1 back to 0.48).
4. **WIDS is in the data, unused by the tree** (lift ≈ 0).
5. **Benign-load control**: unlabeled SYN on wlan0 drops heap $\sim 698xx \rightarrow 681xx$; Phase A DT flags $\sim 98\%$ of those windows as attack (label was 0).
6. We treat (3)–(5) as **findings** — next: **matched-load NORMAL** / feature work, not a new model chase.

What we are building

Edge window IDS on ESP32 (not a per-packet firewall):

- **Sense:** promiscuous 802.11 + on-device host state
- **Decide:** shallow DT every 100 ms (`model.h`)
- **Act (later):** HIPS — **off** for now (FPR still high)

Modality	Role	Phase A status
NIDS	RF / frame counts, RSSI/SNR	Used (secondary)
WIDS	<code>deauth_targeted</code> , <code>seq_jump</code>	In CSV; tree ignores
HIDS	heap, queues, reconnects...	Dominates via <code>heap</code>

Expr A

Capture 20260805 · 6575 windows · balance **PASS**

NORMAL 3746 / DEAUTH 1472 / SYN 645 / ARP 712

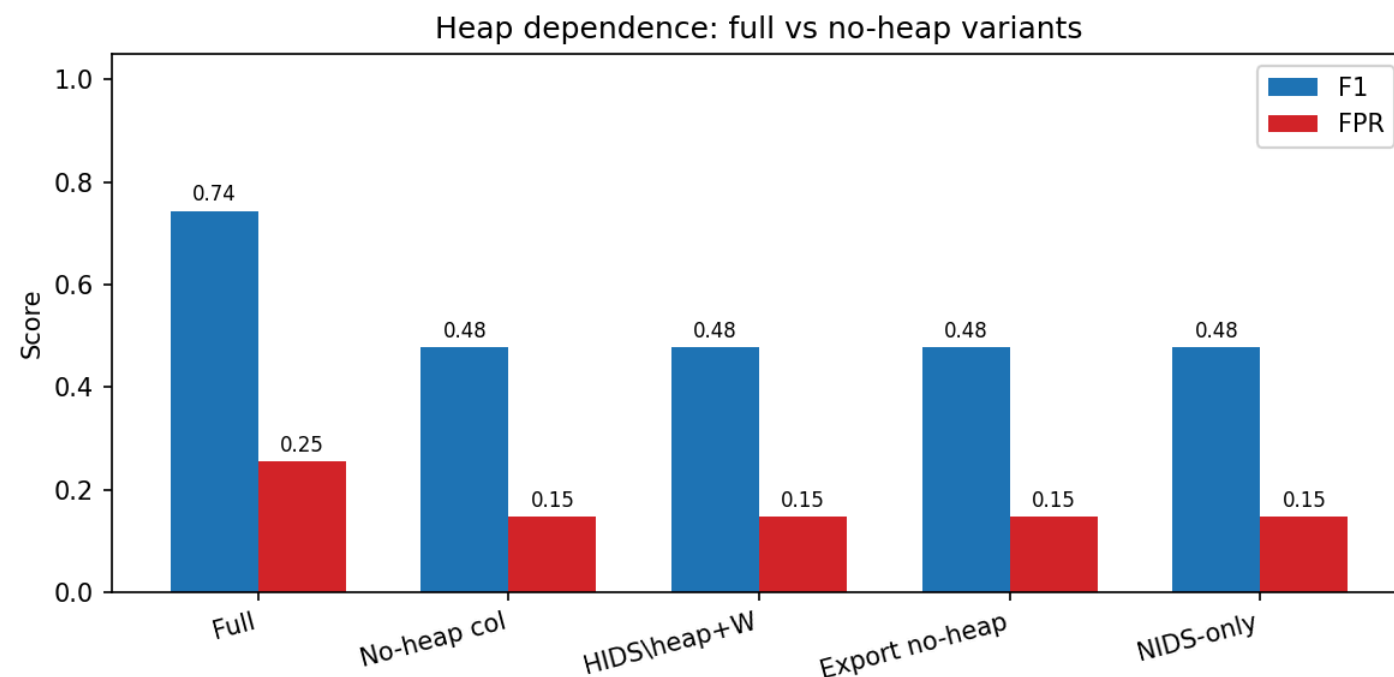
	F1	FPR	AUC
DT (export)	0.743	0.254	0.850
RF	0.717	0.170	0.848

Per-attack recall (DT): SYN 0.95 · ARP **0.97** · DEAUTH 0.62

ROC / CM in backup if needed.

Ablations that matter

Question	Result
Does HIDS help?	Yes — $\Delta F1$ +0.27 vs NIDS-only (0.48)
Does P0 WIDS help?	No — $\Delta F1 \approx 0$
Is HIDS = more than heap?	No — drop / neutralize heap $\rightarrow$ F1 0.48 again



Why heap dominates (interpretation)

`heap` = `esp_get_free_heap_size()` — free RAM at window end.

- Phase A: SYN/ARP $\approx$ 65.8 KB; NORMAL/DEAUTH often $\approx$ 69.8 KB
- Tree learns *low heap* $\rightarrow$ *attack*; hold-out FPR 0.25

Control (unlabeled SYN, no START/STOP):

Segment	heap median	Phase A DT $\rightarrow$ attack
IDLE	$\sim$ 698xx	$\sim$ 25–47%
Unlabeled SYN (strong)	$\sim$ 681xx	$\sim$ 98%
IDLE after	$\sim$ 698xx	elevated, then recovers

- ICMP / wrong-iface UDP did **not** do this; SYN-shaped load did.
- Stronger flood still plateaued $\sim$ **681xx** (not Phase A's 65.8 KB).
- HIDS today $\approx$ **stress / load proxy**, not attack semantics.

Proposed next

Status	What
Done	Benign-load control (heap + FP under unlabeled SYN)
Next	Matched-load NORMAL / interleaved capture → re-train + heap ablation
Defer	New models, HIPS on, no-heap flash

Ask

1. Next capture = **matched-load NORMAL** (background load labeled 0, interleaved with attacks)?
2. Keep WIDS P0 as “instrumented, not yet discriminative,” or redesign in parallel?

Recommend: matched-load recollect → same train + heap ablation.

[Backup] Lab sketch

```
Kali  -- air --> ESP32 (STA + promiscuous)
Kali  -- START/STOP --> Collector
ESP32 -- syslog --> Collector → train → model.h
```

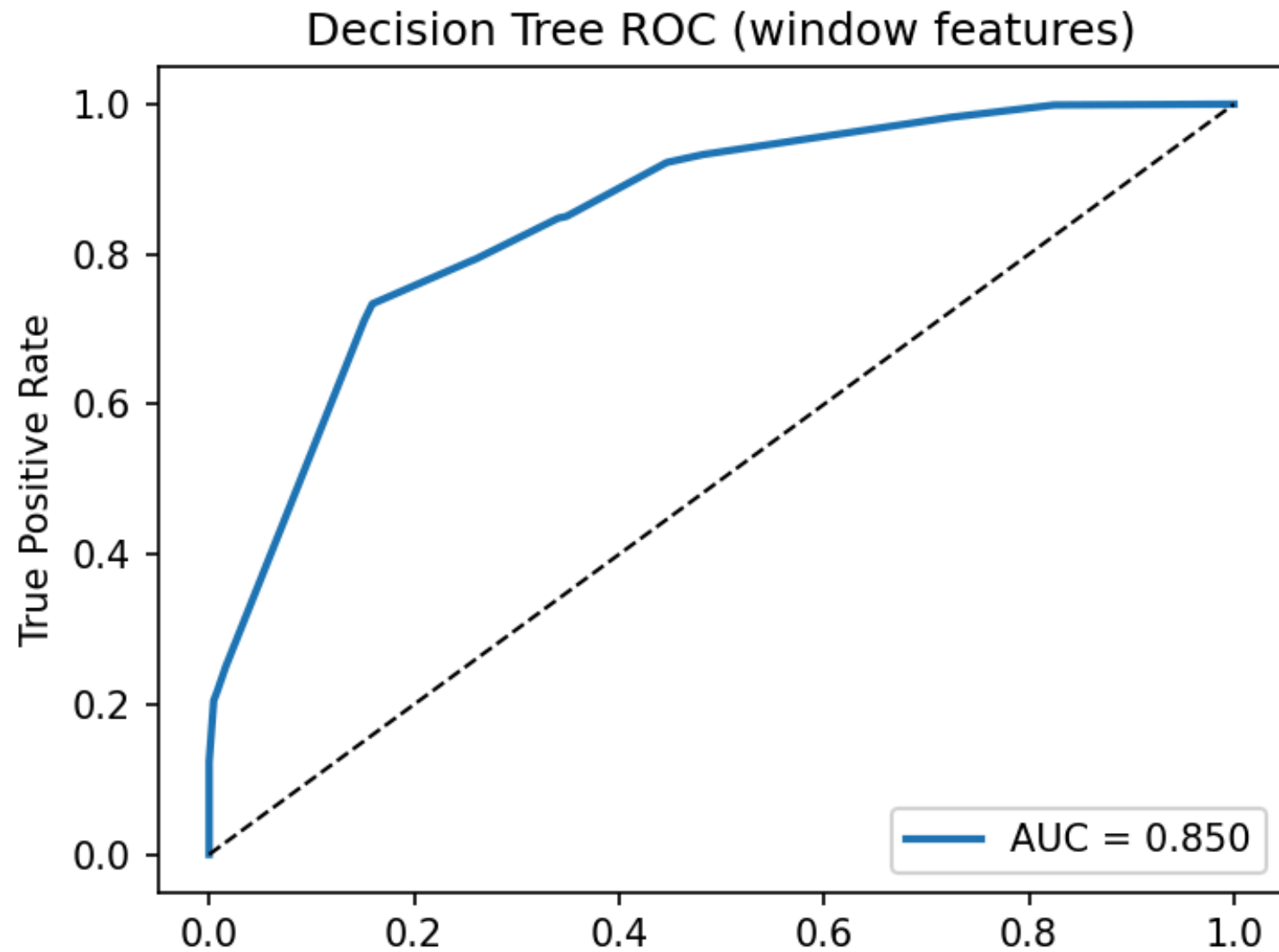
Attacks: deauth · SYN flood · ARP spoof · 100 ms windows · HIPS off while collecting.

[Backup] Feature list (if asked)

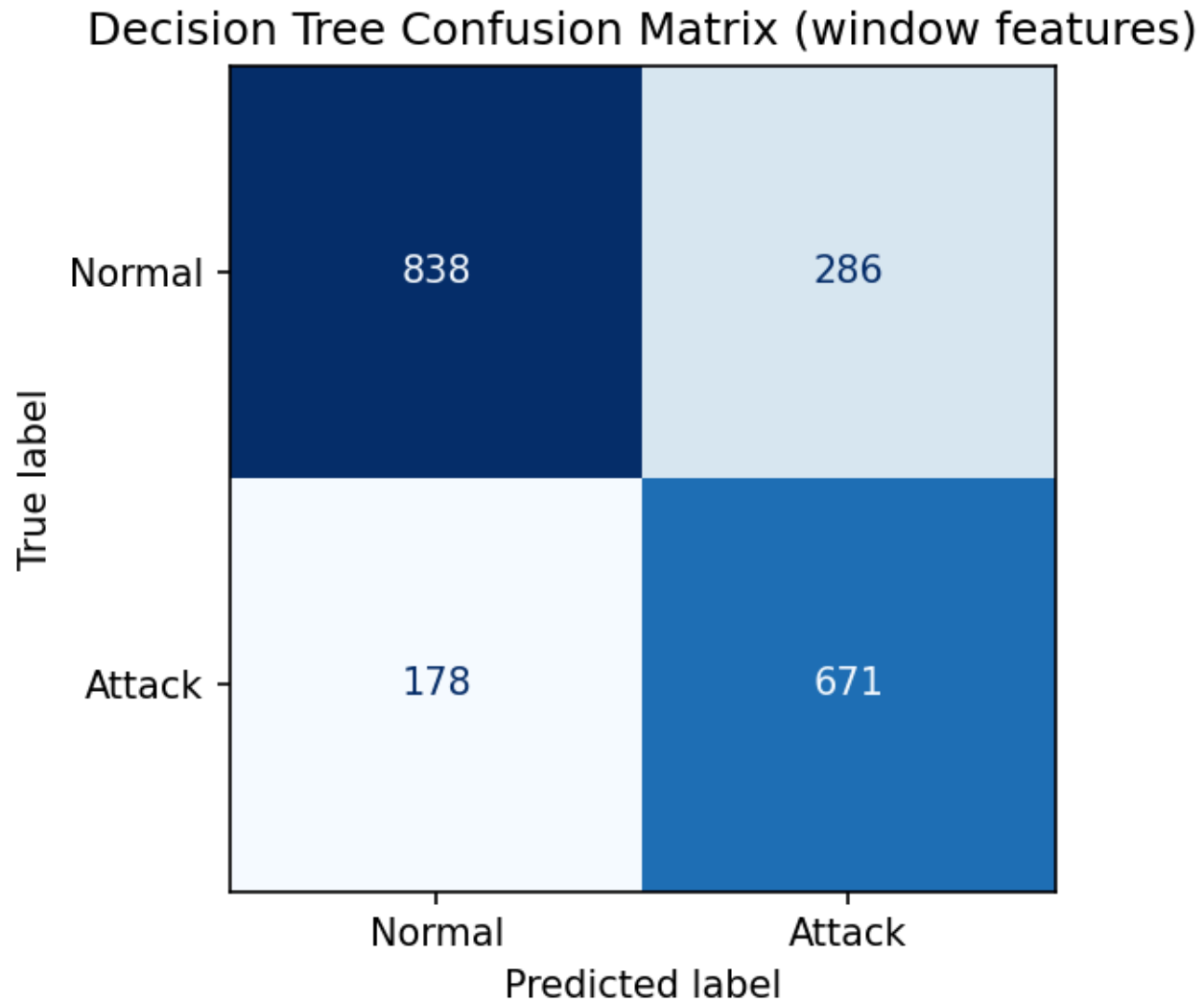
	Features
NIDS	total_packets , packet_density , beacon/deauth/probe/auth , rssi_* , snr_mean
WIDS P0	deauth_targeted (to us / broadcast), seq_jump (seq discontinuity count)
HIDS	heap , minheap , reconn , qpeak , udpfail , backlog

Why WIDS unused this run: seq_jump noisy on mixed MAC traffic; deauth_targeted sparse (~11% of DEAUTH windows). **Do not delete yet** — keep collecting; redesign later.

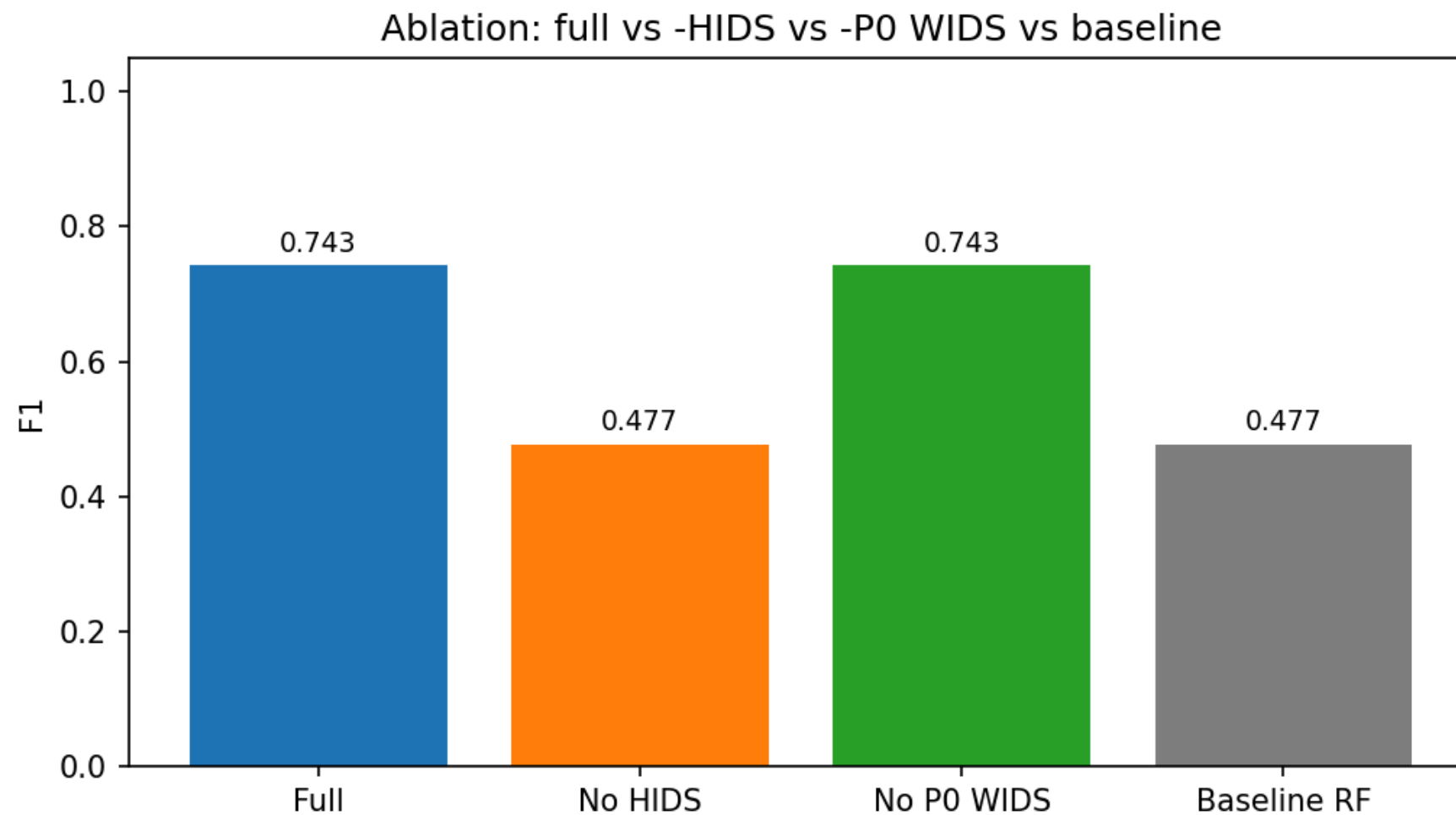
[Backup] ROC



[Backup] Confusion matrix

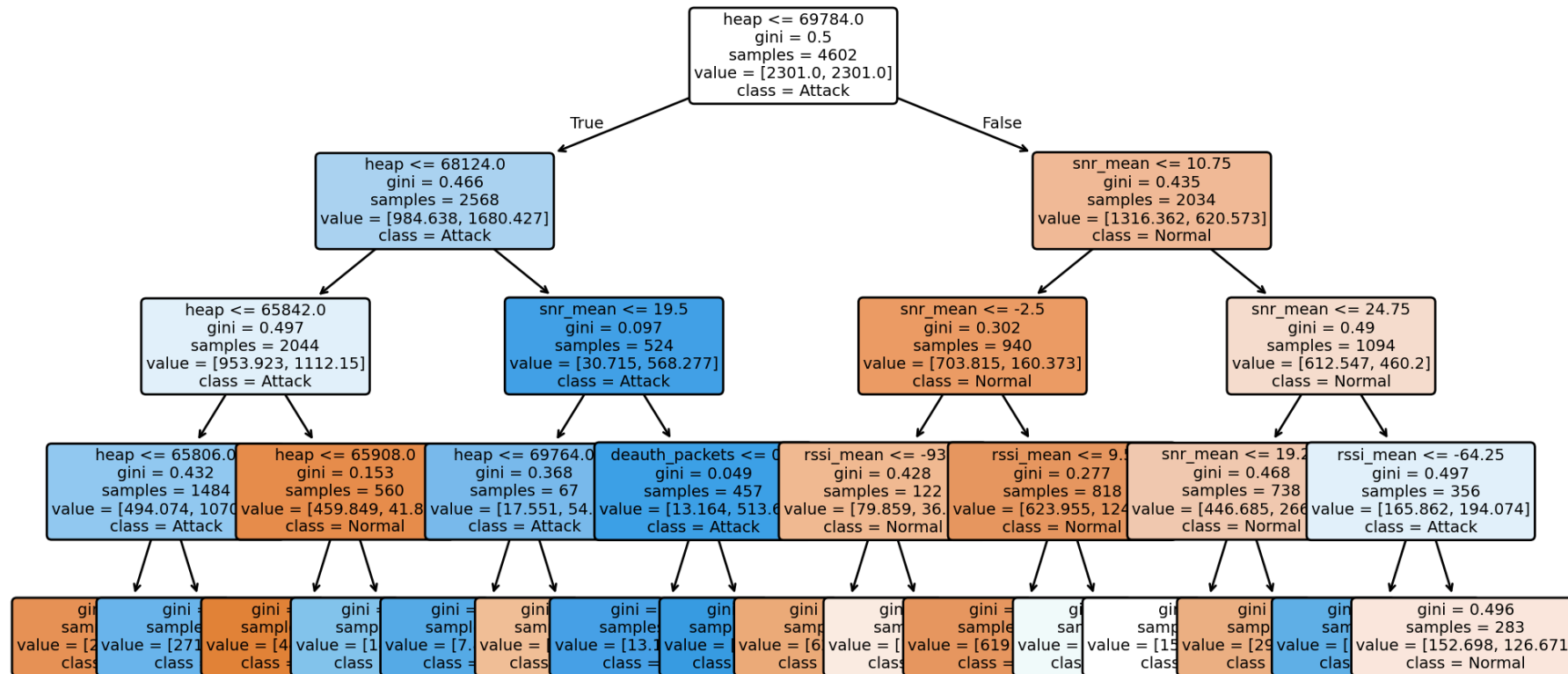


[Backup] Fusion ablation (figure)



[Backup] Exported tree (structure)

Exported Decision Tree (DecisionTree)



[Backup] Artifacts

Path	Content
docs/figures/eval_metrics.json	Full metrics
docs/figures/*.png	Plots
data/windows/nids_windows_20260805_003226.csv	Windows (local)
main/model.h	On-device export (full tree; HIPS off)